Advanced readings and videos

Advanced Readings

- Complex numbers (html)
- Logarithms and Decibels (html)
- Euler's Identity (html)
- Sinusoids and Exponentials (html)
- Geometric Signal Theory (html)

Lecture Videos for Advanced Reading

- Intro to the Fourier Transform (FT) and the Discrete Fourier Transform (DFT)
- DFT as a Matrix Multiply (or N Inner Products)
- Euler's Identity, Complex Sinusoids
- Complex Plane Intro
- Euler's Identity Corollaries
- Euler's Identity Proof
- Sinusoid Demos in Matlab
- DFT Overview, Euler's Identity on the Complex Plane, Additive Synthesis of a Square Wave by Fourier's Theorem
- Sinusoidal Motion as Projection of Circular Motion
- Sinusoidal Motion as Projection of Circular Motion, Continued
- Mathematics of Sinusoids, Complex Sinusoids
- Spectra of Sinusoids
- Amplitude Modulation (AM) and its Spectral Effects, AM Demo, ``Ring Modulation,'' Perception of AM
- Web Resources regarding Hearing, Basilar Membrane
- Circular Motion Projection and Superposition

- Time Constant of Decay, Plotting Exponentials
- Euler's Identity in the Complex Plane
- Generalized Complex Sinusoids
- Generalized Complex Sinusoids II
- Demo Exponentially Decaying Sinusoids in Matlab, Spectrograms, DTMFs
- Making a sinusoid in Pure Data (Pd)
- Exponentials
- AM Review, Frequency Modulation (FM)
- FM Spectra, Bessel Functions
- Generalized Complex Sinusoids Review, The S-Plane
- Laplace Transforms
- Sampled Generalized Complex Sinusoids
- Sampled Generalized Complex Sinusoids II, Z Transforms
- Domain of Z Transforms, The Sampled Sinusoid j^n, Normalized Frequency
- Geometric Signal Theory
- Preview of Orthogonal Projection

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